EXAM 2

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Question 1

1 pt 1

Normal Z table

For a standard normal distribution, find:

$P(z < c) = 0.9001$

Find c rounded to two decimal places.

1.28

0.9001 round down

0.8997

$0.9001 - 0.8997 = 0.0004$

c score = 1.28

Question 2

1 pt 1

For a confidence level of 90%, find the critical value

1.645

$\alpha = 1 - 0.90 = 0.10$

$\frac{\alpha}{2} = \frac{0.10}{2} = 0.05$

Total area to left = $0.90 + 0.05 = 0.95$

use this

$0.9495 < 0.95 < 0.9505$

zscore = 1.645 1.65

Question 3

5 pts 1

The numbers of false fire alarms were counted each month at a number of sites. The results are given in the following table.

Month	Number of Alarms
January	44
February	43
March	33
April	41
May	35
June	26
July	37
August	38
September	26
October	29
November	44
December	43

Test the hypothesis that false alarms are equally likely to occur in any month. Use 5% level of significance.

Procedure: Select an answer ← Goodness of Fit Test

Assumptions: (select everything that applies)

- [x] At most 20% of the expected frequencies are less than 5
- [ ] Independent samples
- [ ] Equal population standard deviations
- [ ] Normal populations
- [x] All expected frequencies are 1 or greater
- [x] Simple random sample

Part 3:

Feb: $43 - 36.5833 = 6.4167 \leftarrow (O - E)$

March: $(-3.5833)^2 = 12.8399 \leftarrow (O-E)^2$

April: $\frac{(O-E)^2}{E} = \frac{19.5072}{36.5833} = 0.5332$

Part 4:

Left CV = DNE

Right CV = $\chi^2 \rightarrow$ df and α

= 19.675

Pvalue = 0.228

According to the CDC, the average weight of men in 1990 was 181 pounds.

Consider a claim that the average weight of men increased by 2014.

(https://abcnews.go.com/Health/average-weight-american-men-15-pounds-20-years/story?id=41100782)

Use these hypotheses:

H0: mean = 181

HA: mean > 181

If you fail to reject the null hypothesis, then your conclusion would be:

- ○ There is sufficient evidence to warrant rejection of the null hypothesis and to support the claim that the average weight of men increased.
- ● There is not sufficient evidence to warrant rejection of the null hypothesis that the average weight of men is the same.

Question 5 1 pt 1

You are conducting a study to see if the average male life expectancy is significantly different from 96. If your null and alternative hypothesis are:

$H_0: \mu = 96$
$H_1: \mu \neq 96$

Then the test is:

- ○ right tailed
- ○ left tailed
- ● two tailed

Question 6 1 pt 1

A statistics instructor believes that fewer than 20% of Evergreen Valley College (EVC) students attended the opening night midnight showing of the latest Harry Potter movie. She surveys 84 of her students and finds that 11 attended the midnight showing. An appropriate alternative hypothesis is:

- ○ p = 0.20
- ○ p ≤ 0.20
- ● p < 0.20
- ○ p > 0.20

Question 7

1 pt 1

You are conducting a study to see if the typical doctor's salary (in thousands of dollars) is significantly less than 92. If your null and alternative hypothesis are:

`H_0: mu = 92`
`H_1: mu < 92`

Then the test is:

- right tailed
- two tailed
- left tailed

Question 8

1 pt 1

You are conducting a study to see if the probability of a true negative on a test for a certain cancer is significantly different from 0.39. You use a significance level of `alpha = 0.10`.

$SE = \sqrt{\frac{p_0(1-p_0)}{n}} \rightarrow \sqrt{\frac{(0.39 \times 0.61)}{685}}$

0.01864

$\hat{p} = \frac{x}{n} = \frac{245}{685} = 0.3577$

`H_0 : p = 0.39`
`H_1 : p != 0.39`

You obtain a sample of size `n=685` in which there are 245 successes.

$z = \frac{\hat{p} - p_0}{SE} = \frac{0.3577 - 0.39}{0.01864} = -1.735$

What is the test statistic for this sample? (Report answer accurate to three decimal places.)

test statistic = -1.735

$P_{value} = 2(0.04135) = 0.0827$

What is the p-value for this sample? (Report answer accurate to four decimal places.)

p-value = 0.0827

The p-value is...

- [x] less than (or equal to) `alpha`
- [ ] greater than `alpha`

This test statistic leads to a decision to...

- [x] reject the null
- [ ] accept the null
- [ ] fail to reject the null

As such, the final conclusion is that...

- [ ] There is sufficient evidence to warrant rejection of the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.
- [ ] There is not sufficient evidence to warrant rejection of the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.
- [x] The sample data support the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.
- [ ] There is not sufficient sample evidence to support the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.

Question 9

1 pt 1

If the probability (p-value) is greater than the level of significance, what decision is made in the hypothesis test?

- [ ] Accept the null hypothesis as true
- [ ] Reject the null hypothesis
- [x] Fail to reject the null hypothesis

Question 10

1 pt 1

In a hypothesis test, a researcher obtains a p-value= 0.5814. The level of significance= 0.001. Give the correct conclusion.

- Fail to reject the null hypothesis
- Accept the alternative hypothesis
- Reject the null hypothesis

Question 11

1 pt 1

For a standard normal distribution, find: $P(-2.23 < z < -0.75)$

$P(z < -0.75) - P(z < -2.23)$

$P(-2.23 < z < -0.75)$

$z = -0.75 \approx 0.2266$

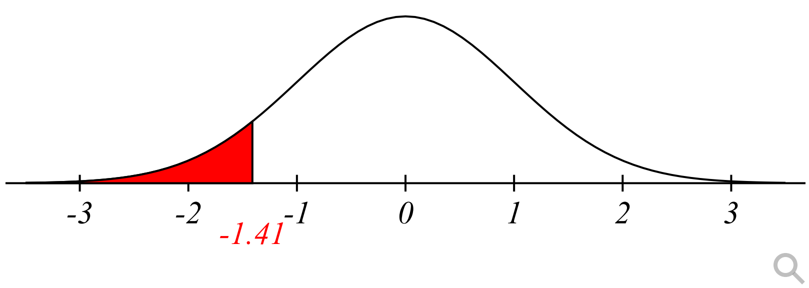
$z = -2.23 \approx 0.0129$ $\longrightarrow$ $0.2266 - 0.0129 = 0.2137$

0.2137

Question 12

1 pt 1

The critical region and critical value for a hypothesis test are shown below. The test statistic is Z = -1.7. Should you reject or fail to reject the null hypothesis.



- Reject
- Fail to Reject

Question 13

1 pt 1

What specifically is used to decide whether to reject or fail to reject the null hypothesis?

- To reject the null hypothesis we need a better alternative hypothesis.
- To reject the null hypothesis we need a sufficiently small p-value.
- To reject the null hypothesis we need a sufficiently large p-value.
- To reject the null hypothesis we need a sufficiently small significance level.

Question 14

1 pt 1

What is the p-value of a hypothesis test?

- A p-value is the probability of finding another random sample with a statistic that is more extreme than the current sample statistic, given that the null hypotheses is wrong.
- A p-value is the probability of finding another random sample with a statistic that is more extreme than the current sample statistic, given that the null hypothesis is correct.
- A p-value is the probability that your sample statistic is less than the population parameter.
- A p-value is a probability that is smaller than the significance level.

Question 15

1 pt 1

How do you tell whether the test is left, right, or two tailed?

- It depends on the alternative hypothesis.
  Left tail for `<`
  Right tail for `>`
  Two tail for `!=`
- It depends on the null hypothesis.
  Left tail for `<=`
  Right tail for `>=`
  Two tail for `=`

Question 16

1 pt 1

You are conducting a study to see if the mean weight of goats (in pounds) is significantly less than 89. Thus you are performing a left-tailed test. The test significance level is `alpha` = .1.

The test statistic, t, has a p-value = 0.14.

Should you reject or fail to reject the null hypothesis? 1

- reject
- fail to reject

Question 17

3 pts 1

The US Department of Energy reported that more than 61% of homes were heated by natural gas. A random sample of 211 homes in Oregon found that 142 were heated by natural gas. Test the claim using a 1% significance level.

a. What type of test will be used in this problem?
Select an answer ← A test for proportion

b. What evidence justifies the use of this test? Check all that apply

- The sample size is larger than 30
- The original population is approximately normal
- The population standard deviation is not known
- There are two different samples being compared
- The population standard deviation is known
- X `np > 5` and `nq > 5`
- The sample standard deviation is not known

c. Enter the null hypothesis for this test.
`H_0` p = 0.61

d. Enter the alternative hypothesis for this test.
`H_1:` p > 0.61

e. Is the original claim located in the null or alternative hypothesis?
Select an answer ← Alt. hypothesis

f. What is the test statistic for the given statistics?
1.876 ← $\hat{p} = \frac{142}{211} = 0.6730$ $SE = \sqrt{\frac{0.61 \cdot 0.39}{211}} = 0.0336 \rightarrow \frac{0.6730 - 0.61}{0.0336} = 1.876$

g. What is the *p*-value for this test?
0.0303 $Z = 1.876 \rightarrow P(Z > 1.876) = 0.0303$

h. What is the decision based on the given statistics?
Select an answer ← Fail to reject the null hypothesis

i. What is the correct interpretation of this decision?

Using a 1 % level of significance, there Select an answer (is not) sufficient evidence to Select an answer (support) the claim that more than 61% of homes were heated by natural gas.

Question 18 5 pts 1

An incubation period is a time between when you contract a virus and when your symptoms start. By surveying randomly selected local hospitals, a researcher was able to obtain the sample of incubation periods of 31 patients. The sample of incubation periods has a mean of 6.06 days and a standard deviation of 1.84 days. Construct a 90% confidence interval for the average incubation period of the novel coronavirus.

i. Procedure: Select an answer ← One mean T procedure

ii. Assumptions: (select everything that applies)

- [x] Simple random sample
- [x] Sample size is greater than 30
- [ ] The number of positive and negative responses are both greater than 10
- [ ] Population standard deviation is known
- [x] Population standard deviation is unknown
- [ ] Normal population

iii. Unknown parameter: Select an answer ← μ, population mean

Question 19 1 pt 1

Part 2

Sample mean $\bar{x} = 6.06$

90% $\alpha = 0.10$

$\frac{a}{2} = 0.05 \rightarrow 1 - \frac{a}{2} = 0.95$

Z score = left -1.697 right 1.697

Part 3:

margin of error = 0.56

Lower band = 5.56

upper band = 6.62

You are conducting a test of homogeneity for the claim that two different populations have the same proportions of the following two characteristics. Here is the sample data.

Category	Population #1	Population #2
A	20	41
B	35	34

→ 61
→ 69
↓ 55 ↓ 75

Tot. (N) = 130

The expected observations for this table would be

Category	Population #1	Population #2
A	25.808	35.192
B	29.192	39.808

$\text{Expected} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$

A P#1 = $\frac{61 \times 55}{130}$ = 25.808
A P#2 = $\frac{61 \times 75}{130}$ = 35.192
BP# 1 = $\frac{69 \times 55}{130}$ = 29.192
BP# 2 = $\frac{69 \times 75}{130}$ = 39.808

The resulting squared Pearson residuals are:

Category	Population #1	Population #2
A	1.307	0.958
B	1.155	0.847

$\text{Residual} = \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$

A P#1 = $\frac{(20-25.808)^2}{25.808}$ = 1.307
A P#2 = $\frac{(41-35.192)^2}{35.192}$ = 0.958
BP#1 = $\frac{(35-29.192)^2}{29.192}$ = 1.155
BP#2 = $\frac{(34-39.808)^2}{39.808}$ = 0.847

1.307 + 0.958 + 1.155 + 0.847 = 4.268

What is the chi-square test-statistic for this data?

\`chi^2 = \` 4.268

Report all answers accurate to three decimal places. (Though you may need to retain more decimal places in the residuals to get the correct chi-square test statistic.)

Hint 1

Question 20 1 pt 1

A soft drink filling machine, when working properly, fills bottles with an average of 12 ounces of soft drink.

If the machine is either over-filling or under-filling, the machine should be shut down and readjusted. You are a quality control employee responsible for monitoring this particular machine and test if the machine works properly.

a. `H_0: mu = 12`.
This null hypothesis implies
Select an answer machine works properly and should not be shut down

b. Select the correct sign for the alternative hypothesis and its meaning. `H_a: mu` ? ≠ `12`.
This alternative hypothesis implies
Select an answer machine does not work properly and should be shut down and readjusted.

c. In this context describe a **Type I error** and the **impact such an error** would have on the company.

- ○ When the machine does not work properly it is not shut down and drinks not filled properly are manufactured. The customers purchased underfilled or overfilled drinks.
- ● When the machine works properly it is shut down and readjusted. This is an unnecessary loss to the company.

d. In this context describe a **Type II error** and the **impact such an error** would have on the company.

- ● When the machine does not work properly it is not shut down and drinks not filled properly are manufactured. The customers purchased underfilled or overfilled drinks.
- ○ When the machine works properly it is shut down and readjusted. This is an unnecessary loss to the company.

e. The **significance level** is probability of making which type of error?

- ○ Type II Error
- ● Type I Error

Question 21 5 pts 1

For a certain drug, based on standards set by the United States Pharmacopeia (USP) - an official public standards-setting authority for all prescription and over-the-counter medicines and other health care products manufactured or sold in the United States, a standard deviation of capsule weights of less than 2 mg is acceptable. A sample of 15 capsules was taken and the weights are provided below:

100.8	98.6	99.4	100.5	100.1
100.1	100.7	101.2	99.9	99
99.6	100.1	99.5	99.4	99.9

The mean of the data set is 99.92; the sample standard deviation is 0.7; if the normality plot is not provided you may assume that the capsule weights are normally distributed.

Construct a 99% confidence interval for the variance (standard deviation) of all capsule weights.

i. Procedure: Select an answer

ii. Assumptions: (select everything that applies)

- The number of positive and negative responses are both greater than 10
- Simple random sample
- Population standard deviation is known
- Population standard deviation is unknown
- Sample size is greater than 30
- Normal population

iii. Unknown parameter: Select an answer

Question 22

1 pt 1

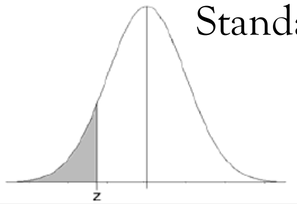
State the indicated conclusion in nontechnical terms, being sure to address the original claim.

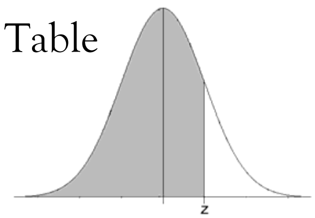
An entomologist writes an article in a scientific journal claiming that **at least** 1.22% of male fireflies are unable to produce light due to a genetic mutation.

A hypothesis test of the claim has been conducted, and the decision from that test is to **reject H_0**.

- There **is** sufficient evidence to **support** the claim that the true proportion is at least 1.22%.
- There is **not** sufficient evidence to warrant **rejection** of the claim that the true proportion is at least 1.22%.
- There **is** sufficient evidence to warrant **rejection** of the claim that the true proportion is at least 1.22%.
- There is **not** sufficient evidence to **support** the claim that the true proportion is at least 1.22%.

Standard Normal Distribution Probabilities Table





z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Confidence Interval Critical Values, $z_{\alpha/2}$

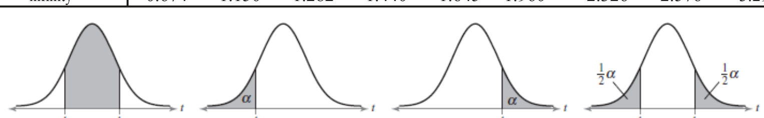
Level of Confidence	Critical Value, $z_{\alpha/2}$
0.90 or 90%	1.645
0.95 or 95%	1.96
0.98 or 98%	2.33
0.99 or 99%	2.575

Hypothesis Testing Critical Values

Level of Significance, α	Left-Tailed	Right-Tailed	Two-Tailed
0.10	− 1.28	1.28	±1.645
0.05	− 1.645	1.645	±1.96
0.01	− 2.33	2.33	±2.575

Student t Distribution Probabilities Table

one-tail area		0.25	0.125	0.1	0.075	0.05	0.025	0.01	0.005	0.0005
two-tail area		0.5	0.25	0.2	0.15	0.1	0.05	0.02	0.01	0.001
confidence level		0.5	0.75	0.8	0.85	0.9	0.95	0.98	0.99	0.999
d.f.	1	1.000	2.414	3.078	4.165	6.314	12.706	31.821	63.657	636.619
	2	0.816	1.604	1.886	2.282	2.920	4.303	6.965	9.925	31.599
	3	0.765	1.423	1.638	1.924	2.353	3.182	4.541	5.841	12.924
	4	0.741	1.344	1.533	1.778	2.132	2.776	3.747	4.604	8.610
	5	0.727	1.301	1.476	1.699	2.015	2.571	3.365	4.032	6.869
	6	0.718	1.273	1.440	1.650	1.943	2.447	3.143	3.707	5.959
	7	0.711	1.254	1.415	1.617	1.895	2.365	2.998	3.499	5.408
	8	0.706	1.240	1.397	1.592	1.860	2.306	2.896	3.355	5.041
	9	0.703	1.230	1.383	1.574	1.833	2.262	2.821	3.250	4.781
	10	0.700	1.221	1.372	1.559	1.812	2.228	2.764	3.169	4.587
	11	0.697	1.214	1.363	1.548	1.796	2.201	2.718	3.106	4.437
	12	0.695	1.209	1.356	1.538	1.782	2.179	2.681	3.055	4.318
	13	0.694	1.204	1.350	1.530	1.771	2.160	2.650	3.012	4.221
	14	0.692	1.200	1.345	1.523	1.761	2.145	2.624	2.977	4.140
	15	0.691	1.197	1.341	1.517	1.753	2.131	2.602	2.947	4.073
	16	0.690	1.194	1.337	1.512	1.746	2.120	2.583	2.921	4.015
	17	0.689	1.191	1.333	1.508	1.740	2.110	2.567	2.898	3.965
	18	0.688	1.189	1.330	1.504	1.734	2.101	2.552	2.878	3.922
	19	0.688	1.187	1.328	1.500	1.729	2.093	2.539	2.861	3.883
	20	0.687	1.185	1.325	1.497	1.725	2.086	2.528	2.845	3.850
	21	0.686	1.183	1.323	1.494	1.721	2.080	2.518	2.831	3.819
	22	0.686	1.182	1.321	1.492	1.717	2.074	2.508	2.819	3.792
	23	0.685	1.180	1.319	1.489	1.714	2.069	2.500	2.807	3.768
	24	0.685	1.179	1.318	1.487	1.711	2.064	2.492	2.797	3.745
	25	0.684	1.198	1.316	1.485	1.708	2.060	2.485	2.787	3.725
	26	0.684	1.177	1.315	1.483	1.706	2.056	2.479	2.779	3.707
	27	0.684	1.176	1.314	1.482	1.703	2.052	2.473	2.771	3.690
	28	0.683	1.175	1.313	1.480	1.701	2.048	2.467	2.763	3.674
	29	0.683	1.174	1.311	1.479	1.699	2.045	2.462	2.756	3.659
	30	0.683	1.173	1.310	1.477	1.697	2.042	2.457	2.750	3.646
	35	0.682	1.170	1.306	1.472	1.690	2.030	2.438	2.724	3.591
	40	0.681	1.167	1.303	1.468	1.684	2.021	2.423	2.704	3.551
	45	0.680	1.165	1.301	1.465	1.679	2.014	2.412	2.690	3.520
	50	0.679	1.164	1.299	1.462	1.676	2.009	2.403	2.678	3.496
	60	0.679	1.162	1.296	1.458	1.671	2.000	2.390	2.660	3.460
	70	0.678	1.160	1.294	1.456	1.667	1.994	2.381	2.648	3.435
	80	0.678	1.159	1.292	1.453	1.664	1.990	2.374	2.639	3.416
	100	0.677	1.157	1.290	1.451	1.660	1.984	2.364	2.626	3.390
	500	0.675	1.152	1.283	1.442	1.648	1.965	2.334	2.586	3.310
	1000	0.675	1.151	1.282	1.441	1.646	1.962	2.330	2.581	3.300
infinity		0.674	1.150	1.282	1.440	1.645	1.960	2.326	2.576	3.291



c-confidence interval — Left-tailed test — Right-tailed test — Two-tailed test

Chi Squared (χ^2) Distribution Probabilities

Area to the Right of Critical Value

d.f.	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169